

Acids, Bases and Salts

Science — Chemistry · Class 7 · Max Marks: 17 · Time: 45 min

Section A — Multiple Choice (Higher Order) (1 × 14 = 14)

1. To tell apart an acid, a base and a neutral solution with certainty, the least you need is:

- A. only turmeric
- B. both red and blue litmus
- C. only blue litmus
- D. no indicator at all

2. A solution turns china rose dark pink and also turns blue litmus red. The solution is:

- A. acidic
- B. both acidic and basic
- C. neutral
- D. basic

3. Acid and base of equal strength are mixed till litmus shows no change. If a little more acid is then added, the mixture will now turn:

- A. china rose green
- B. no litmus at all
- C. red litmus blue
- D. blue litmus red

4. Bubbles of gas form when a pinch of baking soda is dropped into lemon juice. This shows the baking soda:

- A. is completely neutral
- B. is only a colouring
- C. reacts with (neutralises) the acid
- D. is itself an acid

5. Which everyday problem is NOT solved by neutralisation?

- A. boiling water for tea
- B. relieving an ant sting
- C. curing acidity/indigestion
- D. making acidic soil fit for crops

6. Four liquids are tested with turmeric paper; only one turns it red. That liquid is most likely:

- A. orange juice
- B. a window-cleaner containing ammonia
- C. vinegar
- D. lemon juice

7. Marble is basic. A marble statue exposed to acid rain for many years will slowly be:

- A. left unchanged
- B. eaten away (neutralised)
- C. made larger
- D. turned blue

8. A wasp's sting is basic (unlike an ant's acidic sting). The best household remedy for a wasp sting is a mild:

- A. acid such as vinegar
- B. neutral liquid such as water
- C. salt solution
- D. base such as baking soda

9. A soil sample turns china rose indicator green. To make it neutral for most crops, one should mix in:

- A. more lime
- B. decaying leaves or manure (acidic organic matter)
- C. common salt
- D. sand

10. Which statement is ALWAYS true?

- A. Neutral solutions turn litmus red
- B. All salts are acidic
- C. All bases turn red litmus blue
- D. All acids are liquids

11. Two colourless solutions are mixed; the beaker becomes warm and the mixture then has no effect on litmus. The two solutions were probably:

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|--------------|--------------------------|
| A. two bases | B. two neutral solutions |
| C. two acids | D. an acid and a base |

12. The sour taste of an unripe mango and the sting of a nettle leaf are both caused by:

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|-----------------------|----------|
| A. neutral substances | B. bases |
| C. acids | D. salts |

13. Antacids use magnesium or aluminium hydroxide rather than sodium hydroxide because sodium hydroxide is:

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|---|-----------------------|
| A. a strong, corrosive base unsafe to swallow | B. completely neutral |
| C. tasteless | D. actually an acid |

14. When hydrochloric acid is neutralised by sodium hydroxide, the salt formed is:

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|-----------------------|----------------------------------|
| A. ammonium chloride | B. calcium carbonate |
| C. magnesium sulphate | D. sodium chloride (common salt) |

Section B — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** Baking soda is rubbed on an ant's sting.

Reason (R): The ant injects an acidic liquid, which the basic baking soda neutralises.

2. **Assertion (A):** Distilled water does not change the colour of litmus.

Reason (R): Distilled water is neutral.

3. **Assertion (A):** Curd is not usually stored in brass or copper vessels.

Reason (R): The lactic acid in curd reacts with these metals.

Answer Key

Multiple Choice (Higher Order)

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|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. D | 4. C | 5. A |
| 6. B | 7. B | 8. A | 9. B | 10. C |
| 11. D | 12. C | 13. A | 14. D | |

Assertion–Reason

1. A
2. A
3. A